

ECM Summary from RW4 Primary Chain

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely $I(1)$ -like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

RW4 domestic vertical chain is FarmGateUA -> ProducerUA -> ProZorro -> Retail, estimated with both forward and reverse-flow pairs. Farm-gate enters from two alternative reconstruction workbooks and both linear/pchip interpolation variants are carried end-to-end. Retail_combined is the anchored downstream index built from Silpo effective prices, Novus observed prices, and a level-aligned ConsumerUA anchor, while Retail_combined_core keeps the strict retailer-only overlap for robustness.

family=ECM

rows=286

source=/Users/getapple/Documents/KSE/Master Thesis/Main materials/Model/Charniuk_Dairy_Research/outputs/primary_chain_summary/primary_chain_consolidated.xlsx

Table: ECM_Summary

panel_level	panel_name	segment_name	product	standardized_type	retailer_panel	brand	price_variant	reconstruction_variant	farm_gate_source	frequency	intersection_rule	pair_scope	chain_direction	stage_from	stage_to	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	unreliable_reason	shock_dummy_included	promote_controls_included	ljungbox_p	arch_p	jb_p	notes	robust_linear_vs_pchip	interpolation_sensitive	robust_across_reconstruction	core_finding_flag
pairwise_product	pairwise::farm_gate_produ...	Вершки	Вершки	cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	forward	FarmGateUA	ProducerUA	farmgateua_to_producerua	ECM	ProducerUA	FarmGateUA	1453	0.5046611574664945	1.197631175348907	-0.008222170527985115	2.231873877761981e-07	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	3.430067571019925e-48	2.424591070955723e-36	0.0	engle_granger_p=1.879e-11...	1	0	0	0
pairwise_product	pairwise::farm_gate_produ...	Вершки	Вершки	cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	reverse	ProducerUA	FarmGateUA	producerua_to_farmgateua	ECM	FarmGateUA	ProducerUA	1458	0.04522135861906845	0.667752610852099	-0.001936823931610217	0.05809863088196516	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	1.05840165009701e-07	9.974202457880823e-40	0.0	engle_granger_p=2.706e-07...	1	0	0	0
pairwise_product	pairwise::farm_gate_produ...	Кефір	Кефір	milk	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	forward	FarmGateUA	ProducerUA	farmgateua_to_producerua	ECM	ProducerUA	FarmGateUA	1458	0.434314987983099	0.7583304594456308	-0.01419289552195011	0.109090290745603e-14	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	5.373143546057728e-51	5.501893668604891e-47	0.0	engle_granger_p=4.9126e-13...	1	0	0	0
pairwise_product	pairwise::farm_gate_produ...	Кефір	Кефір	milk	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	reverse	ProducerUA	FarmGateUA	producerua_to_farmgateua	ECM	FarmGateUA	ProducerUA	1458	0.03205360642959576	0.8310673931365553	-0.001100288103428076	0.144363189817243	nan	nan	nan	unreliable	1	residual_autocorr_or_arch...	0	0	2.801043471169433e-09	2.243172233068431e-44	0.0	engle_granger_p=1.713e-05...	1	0	0	0
pairwise_product	pairwise::farm_gate_produ...	Молоко питне	Молоко питне	milk	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	forward	FarmGateUA	ProducerUA	farmgateua_to_producerua	ECM	ProducerUA	FarmGateUA	1458	0.3006872990624536	0.285205834402301	-0.01228863024179999	4.284114762962924e-14	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	4.304350654312129e-53	1.320665998920487e-46	0.0	engle_granger_p=3.202e-13...	1	0	0	0
pairwise_product	pairwise::farm_gate_produ...	Сир твердий	Сир твердий	hard_cheese	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	forward	FarmGateUA	ProducerUA	farmgateua_to_producerua	ECM	ProducerUA	FarmGateUA	1453	0.441975075223305	0.6423572673578484	-0.009890272876371701	2.869045501039659e-08	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	1.80534626350597e-52	1.054990591795885e-39	0.0	engle_granger_p=1.546e-15...	1	0	0	0
pairwise_product	pairwise::farm_gate_produ...	Сир твердий	Сир твердий	hard_cheese	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	reverse	ProducerUA	FarmGateUA	producerua_to_farmgateua	ECM	FarmGateUA	ProducerUA	1458	0.03728545116104544	0.8808404259202449	-0.0008805113744489553	0.1788100780055836	nan	nan	nan	unreliable	1	residual_autocorr_or_arch...	0	0	1.566088974913432e-09	2.2744169876774786e-41	0.0	engle_granger_p=2.685e-05...	1	0	0	0
pairwise_product	pairwise::farm_gate_produ...	Сметана	Сметана	sour_cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	forward	FarmGateUA	ProducerUA	farmgateua_to_producerua	ECM	ProducerUA	FarmGateUA	1458	0.4767282371653001	0.7898248144586324	-0.01277757950384819	4.51055195483249e-13	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	7.818545527548611e-49	9.31999246045849e-40	0.0	engle_granger_p=1.221e-12...	1	0	0	0
pairwise_product	pairwise::farm_gate_produ...	Сметана	Сметана	sour_cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	reverse	ProducerUA	FarmGateUA	producerua_to_farmgateua	ECM	FarmGateUA	ProducerUA	1458	0.03733473255883056	0.8080122046707814	-0.00105177745776355	0.1747417791225999	nan	nan	nan	unreliable	1	residual_autocorr_or_arch...	0	0	1.793462841862316e-08	3.018365846872249e-42	0.0	engle_granger_p=1.679e-05...	1	0	0	0
product	product::Сир твердий::Sil...	Сир твердий	Сир твердий	hard_cheese	Silpo	nan	observed	linear	initial	daily	chain_common_support	full_chain	reverse	Retail	FarmGateUA	retail_to_farmgateua	ECM	FarmGateUA	Retail	36	-0.0006243607074365656	-0.05118671340979519	-0.1304047954777818	0.01618224692205509	nan	nan	nan	ok	0	nan	0	0	0.45465911567329976	0.1381724793310779	0.03583439982485814	engle_granger_p=0.02972; ...	0	1	0	0
pairwise_product	pairwise::farm_gate_retail...	Сир кисломолочний	Сир кисломолочний	cottage_cheese	Silpo	nan	observed	linear	initial	daily	pairwise_overlap	farm_gate_retail	forward	FarmGateUA	Retail	farmgateua_to_retail	ECM	Retail	FarmGateUA	46	-1.411185376998817	-0.6668726737989257	-0.7111275008459168	2.337288684082808e-05	nan	nan	nan	ok	0	nan	0	0	0.5781530247944058	0.8908964703188634	0.3331366470316252	engle_granger_p=0.02964; ...	1	0	0	0
pairwise_product	pairwise::farm_gate_retail...	Сир кисломолочний	Сир кисломолочний	cottage_cheese	Silpo	nan	observed	linear	initial	daily	pairwise_overlap	farm_gate_retail	reverse	Retail	FarmGateUA	retail_to_farmgateua	ECM	FarmGateUA	Retail	46	-0.002000336060425197	-0.02068561938699354	-0.06896994761135933	0.05745292963906834	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	0.067225562253118433	9.445715966864688e-21	engle_granger_p=0.02379; ...	1	0	0	0	
pairwise_product	pairwise::prozorro_retail...	Сир твердий	Сир твердий	hard_cheese	Silpo	nan	observed	linear	initial	daily	pairwise_overlap	prozorro_retail	reverse	Retail	FarmGateUA	retail_to_farmgateua	ECM	FarmGateUA	Retail	36	-0.0006243607074365656	-0.05118671340979519	-0.1304047954777818	0.01618224692205509	nan	nan	nan	ok	0	nan	0	0	0.4546591156239976	0.1381724793310779	0.03583439982485814	engle_granger_p=0.02972; ...	1	0	0	0
pairwise_product	pairwise::farm_gate_retail...	Інше/невідомо	Інше/невідомо	other	Silpo	nan	baseline	linear	initial	daily	pairwise_overlap	farm_gate_retail	reverse	Retail	FarmGateUA	retail_to_farmgateua	ECM	FarmGateUA	Retail	52	0.0004284409225518853	0.01067910016805668	-0.06978266038770009	0.0716474864066191	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	0.1462498406236656	0.006674092855700993	4.783657699796243e-27	engle_granger_p=0.01047; ...	1	0	0	0
pairwise_product	pairwise::farm_gate_retail...	Йогурт	Йогурт	yogurt_dessert	Silpo	nan	baseline	linear	initial	daily	pairwise_overlap	farm_gate_retail	forward	FarmGateUA	Retail	farmgateua_to_retail	ECM	Retail	FarmGateUA	46	0.9781715097661041	-1.79344702189178	-0.5187258717432635	0.001385064388329802	nan	nan	nan	ok	0	nan	0	0	0.9579665478671815	0.8249640495523929	0.4317833169843933	engle_granger_p=0.01982; ...	1	0	0	0
pairwise_product	pairwise::farm_gate_retail...	Молоко питне	Молоко питне	milk	Silpo	nan	baseline	linear	initial	daily	pairwise_overlap	farm_gate_retail	forward	FarmGateUA	Retail	farmgateua_to_retail	ECM	Retail	FarmGateUA	46	-1.5615055829480035	-1.687476830915444	-0.7699475155353299	3.21706597430559e-07	nan	nan	nan	ok	0	nan	0	0	0.6549943593397164	0.9071395639198819	0.3916090752767693	engle_granger_p=0.0001217...	1	0	0	0
pairwise_product	pairwise::farm_gate_retail...	Молоко питне	Молоко питне	milk	Silpo	nan	baseline	linear	initial	daily	pairwise_overlap	farm_gate_retail	reverse	Retail	FarmGateUA	retail_to_farmgateua	ECM	FarmGateUA	Retail	46	-0.003671582500940047	-0.1358304275529033	-0.09226667775386316	0.008515684510009997	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	0.1136186287883972	0.001730064502182112	2.246360130432368e-14	engle_granger_p=0.07921; ...	1	0	0	0
pairwise_product	pairwise::producer_retail...	Молоко питне	Молоко питне	milk	Silpo	nan	baseline	linear	initial	daily	pairwise_overlap	producer_retail	forward	FarmGateUA	Retail	farmgateua_to_retail	ECM	Retail	FarmGateUA	46	-1.5615055829480035	-1.687476830915444	-0.7699475155353299	3.21706597430559e-07	nan	nan	nan	ok	0	nan	0	0	0.6549943593397164	0.9071395639198819	0.3916090752767693	engle_granger_p=0.0001217...	1	0	0	0
pairwise_product	pairwise::producer_retail...	Молоко питне	Молоко питне	milk	Silpo	nan	baseline	linear	initial	daily	pairwise_overlap	producer_retail	reverse	Retail	FarmGateUA	retail_to_farmgateua	ECM	FarmGateUA	Retail	46	-0.003671582500940047	-0.1358304275529033	-0.09226667775386316	0.008515684510009997	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	0.1136186287883972	0.001730064502182112	2.246360130432368e-14	engle_granger_p=0.07921; ...	1	0	0	0
pairwise_product	pairwise::farm_gate_retail...	Молоко питне	Молоко питне	milk	Retail_combined	nan	observed	linear	initial	daily	pairwise_overlap	farm_gate_retail	forward	FarmGateUA	Retail	farmgateua_to_retail	ECM	Retail	FarmGateUA	1456	0.5632789134311367	0.7698303569083205	-0.0690933647813763	1.101554973927207e-06	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	3.805482434556599e-07	2.243512686698309e-36	0.0	engle_granger_p=3.727e-09...	1	0	0	0
pairwise_product	pairwise::farm_gate_retail...	Молоко питне	Молоко питне	milk	Retail_combined	nan	observed	linear	initial	daily	pairwise_overlap	farm_gate_retail	reverse	Retail	FarmGateUA	retail_to_farmgateua	ECM	FarmGateUA	Retail	1460	0.0004408421456836772	0.7773202673325867	-0.0006190018574945828	0.4307260548486532	nan	nan	nan	unreliable	1	residual_autocorr_or_arch...	0	0	5.675243349200851e-12	5.781061898842001e-51	0.0	engle_granger_p=0.001696; ...	1	0	0	0
pairwise_product	pairwise::farm_gate_retail...	Сир твердий	Сир твердий	hard_cheese	Retail_combined	nan	observed	linear	initial	daily	pairwise_overlap	farm_gate_retail	forward	FarmGateUA	Retail	farmgateua_to_retail	ECM	Retail	FarmGateUA	1460	0.7499046011888078	0.8617947096067899	-0.05131765958325796	0.0002192948638802699	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	2.940114389548179e-12	8.858278798310606e-97	0.0	engle_granger_p=5.357e-10...	1	0	0	0
pairwise_product	pairwise::farm_gate_retail...	Сир твердий	Сир твердий	hard_cheese	Retail_combined	nan	observed	linear	initial	daily	pairwise_overlap	farm_gate_retail	reverse	Retail	FarmGateUA	retail_to_farmgateua	ECM	FarmGateUA	Retail	1460	0.001254140813331646	0.78467315200020258	-0.0004172910153417971	0.6125776447827794	nan	nan	nan	unreliable	1	residual_autocorr_or_arch...	0	0	9.566601379474948e-12	1.317650735405743e-50	0.0	engle_granger_p=7.117e-05...	1	0	0	0
pairwise_product	pairwise::farm_gate_retail...	Сметана	Сметана	sour_cream	Retail_combined	nan	observed	linear	initial	daily	pairwise_overlap	farm_gate_retail	forward	FarmGateUA	Retail	farmgateua_to_retail	ECM	Retail	FarmGateUA	1455	0.217691618650793	0.9447553571324225	-0.04163979149866366	4.200685384148623e-06	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	3.3110116733727506e-182	1.791716133727506e-182	0.0	engle_granger_p=4.506e-11...	1	0	0	0
pairwise_product	pairwise::farm_gate_retail...	Сметана	Сметана	sour_cream	Retail_combined	nan	observed	linear	initial	daily	pairwise_overlap	farm_gate_retail	reverse	Retail	FarmGateUA	retail_to_farmgateua	ECM	FarmGateUA	Retail	1460	0.001257468119256966	0.7427386921817747	-0.0006722626101732968	0.4458857326042251	nan	nan	nan	unreliable	1	residual_autocorr_or_arch...	0	0	9.4714552124282081e-12	9.780477871379841e-51	0.0	engle_granger_p=0.0001654...	1	0	0	0
pairwise_product	pairwise::producer_retail...	Молоко питне	Молоко питне	milk	Retail_combined	nan	observed	linear	initial	daily	pairwise_overlap	producer_retail	forward	FarmGateUA	ProducerUA	farmgateua_to_producerua	ECM	ProducerUA	FarmGateUA	1458	0.3006872990624536	0.285205834402301	-0.01228863024179999	4.284114762962924e-14	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	4.304350654312129e-53	1.320665998920487e-46	0.0	engle_granger_p=3.202e-13...	1	0	0	0
pairwise_product	pairwise::producer_retail...	Молоко питне	Молоко питне	milk	Retail_combined	nan	observed	linear	initial	daily	pairwise_overlap	producer_retail	forward	FarmGateUA	Retail	farmgateua_to_retail	ECM	Retail	FarmGateUA	1454	0.5949595716991603	0.7701798483451432	-0.06894767944914375	1.129423279445182e-06	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	1.760439756514154e-06	2.490539851624045e-36	0.0	engle_granger_p=1.578e-10...	1	0	0	0
pairwise_product	pairwise::producer_retail...	Молоко питне	Молоко питне	milk	Retail_combined	nan	observed	linear	initial	daily	pairwise_overlap	producer_retail	reverse	Retail	ProducerUA	retail_to_producerua	ECM	ProducerUA	Retail	1786	0.0126124286154411	0.6911881232649273	-0.004568985654472914	0.04937525204822524	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	1.363416386506709e-61	3.145044635227592e-43	0.0	engle_granger_p=0.05307; ...	1	0	0	0
pairwise_product	pairwise::producer_retail...	Молоко питне	Молоко питне	milk	Retail_combined	nan	observed	linear	initial	daily	pairwise_overlap	producer_retail	reverse	Retail	FarmGateUA	retail_to_farmgateua	ECM	FarmGateUA	Retail	1458	0.0004664334944																			